

NEUFLO Syrup 5mg/ml

Active ingredient(s):

Each ml contains:

Lysozyme Chloride 5.0 mg (P)

Preservative:

Each ml contains:

Methyl Paraben (as preservative) 1.8 mg

Propyl Paraben (as preservative) 0.2 mg

CONTRAINDICATIONS

(This product is contraindicated in the following patients)

1. Patients with a history of hypersensitivity to any ingredient of this drug.

2. Patients with allergy to egg-white

[The ingredient of this product is a protein derived from egg-white and symptoms of hypersensitivity including anaphylactic shock have been reported in patients with allergy to egg-white]

Pharmacology (Summary of Pharmacodynamic and Pharmacokinetic):

Pharmacodynamic:

1. Anti-inflammatory effect

Lysozyme hydrochloride inhibits footpad edema (rats) and proliferation of granuloma (rats). Lysozyme hydrochloride also improves increased vascular permeability and congestion caused by thermal burn (rabbit's skin).

2. Effects on scar formation/tissue repair

Lysozyme hydrochloride promotes scarring of rabbit's cuts and inhibits suppuration and necrosis. Lysozyme hydrochloride also promotes repair of tissue damage in an experiment with nodules (rats) that were formed by the administration of absorbent cotton balls and gel-like agar.

3. Decomposition of purulent mucus and excretory effect

Lysozyme hydrochloride decreases the viscosity of pus or sputum in patients with chronic sinusitis of respiratory disease and enhances clearance of secretions by activating ciliary movement of ciliated epithelial cells.

*Clinical Study done for Leftose Syrup 0.5% (contains lysozyme):

Note : The following clinical studies were conducted on the innovator product (Leftose Syrup 0.5% which contains Lysozyme 5.0mg/ml).

- Efficacy on difficulty of expectoration

The efficacy rate ("effective" or better) of Leftose Syrup 0.5% in respiratory diseases presenting with a chief complaint of difficulty of expectoration in the open study was 78% (147/188).

A double-blind study was also conducted in 299 patients including those in the control group and demonstrated the efficacy of Leftose Syrup 0.5% in expectoration compared to the control drug.

- Efficacy on chronic sinusitis

The efficacy rate of Leftose Syrup 0.5% in chronic sinusitis was 52% (24/46)

Pharmacokinetic:

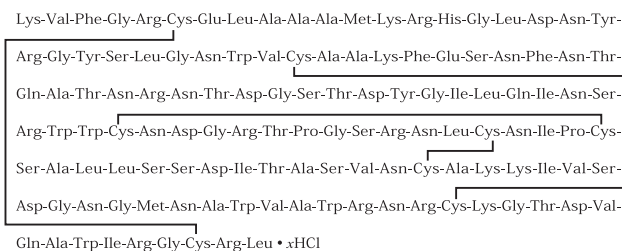
Absorption in animals

When lysozyme hydrochloride was orally given in rabbits, the blood concentration reached to the maximum level in approximately 60 minutes and followed by a gradual decrease in concentrations. Furthermore, gastrointestinal absorption study in rats revealed that lysozyme hydrochloride was mainly absorbed via the portal system.

PHYSICOCHEMISTRY

Nonproprietary name: Lysozyme hydrochloride

Structural formula:



Molecular formula: $C_{616}H_{962}N_{193}O_{182}S_{16} \cdot xHCl$
 Basic polypeptide consisting of 129 amino acids
 of 18 different types

Molecular weight: Approximately 14,500

Description: Lysozyme hydrochloride occurs as white crystals
 or crystalline or amorphous powder.
 It is freely soluble in water and practically insoluble in ethanol (99.5).
 It is hygroscopic.
 The pH of a solution of lysozyme hydrochloride
 (3 in 200) is between 3.0 to 5.0

Indication(s):

Chronic paranasal sinusitis, difficulty of expectoration caused by respiratory diseases.

Dosage and Administration:

Usually, the following dosage are divided in 3 times daily and administered orally.

Adult:

12- 54ml {60-270mg (potency) as lysozyme chloride}

Children (Usual Daily Dose):

<2 years old : 3-6ml {15-30mg(potency) as lysozyme chloride}
 2-6 years old : 6-8 ml {30-40mg(potency) as lysozyme chloride}
 7-14 years old : 8-12 ml {40-60mg(potency) as lysozyme chloride}

In addition, the dose may be adjusted depending on the symptoms. There are still unknown factors concerning action mechanism of this drug in human body, and correlation between its dose and effects has not been fully clarified. It is suggested, therefore, that this drug should not be administered without specific therapeutic purpose.

Mode of Administration:

Neuflo Syrup can be taken with or without food.

Precaution(s) / Warning(s):

1. **Careful administration** (*this product should be administered with care in the following patients*)
 - a. Patients with a predisposition to disease manifested as dermatitis atopic, asthma bronchial, drug allergy and food allergy, etc.
[Patients with predisposition to disease may be sensitive to various allergens including drugs. Anaphylactoid reaction may occur in such patients.]
 - b. Patients with a family history of allergic symptoms in parents, brothers or sisters, etc.
[If an predisposition to disease has been inherited, allergy may occur.]
2. **Important precaution**
It has been reported that anaphylactoid reactions such as shock, etc. occurred in infants during the initial oral dose of this drug, parents of patient should be given sufficient explanation on flushing, urticaria, face edema and dyspnoea, etc. Parents should be instructed to discontinue administration and promptly receive consultation if any symptom is observed.

Interaction with Other Medicaments:

None known.

Pregnancy and Lactation:

As safety of this drug has not been well established, use with caution in pregnant women or suspected of being pregnant.

Side Effect(s):

Note : The following are the possible adverse reactions for lysozyme hydrochloride preparations. The data are based on the approved package insert of the innovator product (Leftose Syrup 0.5% which contains Lysozyme 5.0mg/ml).

In total 770 patients, adverse reactions were observed in 14 patients (1.82%) and most of them were gastrointestinal symptoms (1.56%). Rash was observed in one patient (0.13%).

1. Clinical significant adverse reactions:
 - a. Shock or anaphylactoid reactions:
Shock or anaphylactoid reactions may occur. Patients should be carefully observed and if any symptoms such as pallor facial, cold extremities, blood pressure decreased, cyanosis, loss of consciousness, flushing, urticaria, face oedema, laryngeal oedema and dyspnoea, etc are observed, administration should be discontinued and appropriate measure should be taken.
 - b. Toxic epidermal necrolysis (TEN), oculomucocutaneous syndrome (Stevens-Johnson syndrome):
Toxic epidermal necrolysis or oculomucocutaneous syndrome may occur. Patients should be carefully observed and if any symptoms such as pyrexia, erythema, itching, ocular hyperaemia and stomatitis, etc are observed, administration should be discontinued and appropriate measures should be taken.
2. Other adverse reactions:

Type \ Incidence	≥0.1% and <5%	<0.1%	Incidence unknown
Hypersensitivity ^(Note)		Rash/redness	
Hepatic	Diarrhoea	Nausea/vomiting	Stomatitis, anorexia, stomach discomfort
Other	-	-	Hepatic function disorder (AST (GOT) increased, ALT (GPT) increased, ALP increased, GGTP increased, LDH increased, etc.)
Other	-	-	Dizziness

Note): In the event of such symptoms, administration should be discontinued.

Symptoms and Treatment for Overdosage, and Antidote(s):

There is no specific antidote for its over dosage; therefore, management of the patients should consist of symptomatic and supportive therapy.
Early gastric lavage may be indicated depending on the amount of overdosage and nature of symptoms.

Shelf-Life:

2 years from the date of manufacture.

Storage Condition(s):

Store at temperature below 30°C.

Product Description & Packing(s):

Clear and yellow color solution with banana flavor.
Plastic bottle of 60ml, 100ml & 120ml.



Manufacturer and Product Registration Holder:
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